



Development of an Automated Project and Test Management System using Frappe Framework

Project Documentation Submitted to the Faculty of the
School of Computing and Information Technologies

Asia Pacific College

In Partial Fulfillment of the Requirements for
Project Management

PROJMAN

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2025

Table of Contents

Business Case	3
Executive Summary	3
Issue	3
Anticipated Outcomes	3
Recommendation	4
Justification	4
Business Case Analysis Team	5
Problem Definition	6
Problem Statement	6
Organizational Impact	7
Technology Migration	8
Project Overview	8
Project Description	8
Goals and Objectives	9
Project Assumptions	10
Project Constraints	12
Major Project Milestones	13
Strategic Alignment	13
Cost Benefit Analysis	14
Alternative Analysis	15
Project Charter	16
Project Purpose/Justification	18
Business Need/Case	19
Business Objectives	20
Project Description	21
Project Objectives and Success Criteria	22
Requirements	23
Constraints	25
Assumptions	26
Preliminary Scope Statement	26
Risks	27
Project Deliverables	28
Summary Milestone Schedule	30
Summary Budget	31

Project Approval Requirements	31
Project Manager	32
Stakeholder Management Strategy	33
Introduction	34
Objectives	34
Identify Stakeholders	34
Stakeholders Categories	35
Key Stakeholders	35
Stakeholder Analysis	36
Power Interest/Grid	36
Stakeholder Engagement Plan	37
Communication Plan	37
Risk Mitigation and Response	38
Scope Analysis	38
Introduction	38
Scope Management Approach	39
Roles and Responsibilities	41
Scope Definition	42
Project Scope Statement.....	43
Project Deliverables	43
Project Exclusions	44
Project Constraints	44
Project Assumptions	45
Scope Verification.....	46
Scope Control.....	46
Schedule Management.....	47
Introduction.....	47
Schedule Management Approach.....	47
Scheduling Tool	48
Schedule Milestones	48
Schedule Control	49
Roles and Responsibilities	49
Schedule Changes and Thresholds	50
Scope Change.....	50
Cost Management.....	51
Introduction.....	52
Cost Management Approach.....	52

Measuring Project Costs	53
Reporting Format	53
Cost Variant Response Process	53
Cost Change Control Process	54
Project Budget Estimate	54
Work Breakdown Structure	55
Introduction	55
Outline View	55
Tabular Structure	56
Tree Structure	57
WBS Dictionary	58
Work Packages Based on WBS	59
Human Resource Plan	60
Introduction	60
Roles and Responsibilities	61
Project Organizational Charts	62
Staffing Management	62
Quality Management.....	64
Introduction	65
Quality Management Approach	65
Quality Requirement and Standards	66
Quality Assurance (QA)	66
Quality Control	66
Quality Control Measurements	67
Communication Management	68
Introduction	69
Communication Management Approach	69
Communication Management Constraints	70
Stakeholder Communication Requirements	71
Roles	71
Project Team Directory	72
Communication Methods Technology	73
Communication Matrix	74
Communication Flowchart	75
Guidelines for Meetings	76
Communication Standards	77
Communication Escalation Process	78

Change Management	80
Introduction	81
Change Control Board	81
Roles and Responsibilities	81
Change Control Process	83
Implementation/Transition	85
Executive Summary	85
Transition Team Organization	85
Workforce Transition	85
Work Executing During Transition	85
Property Transition	86
Intellectual Property	86
User Accounts and Passwords	86
Knowledge Transfer	86
Schedule	87
Handover Acceptance	90

Business Case

Executive Summary

Strong and integrated systems are needed in modern organizations to complement the complexities of the project execution process and the quality of the products. Enterprise Resource Planning (ERP) systems such as Frappe are usually the foundation of the core business processes but do not have specialized in-depth features to support project management and quality assurance. Because of this, a number of companies now must choose to use a hybrid model, where they use their ERP to manage their data, but as a means of managing tasks and testing, they use external, third-party software like Jira. This method, nevertheless, poses considerable issues of data fragmentation, cost, and inefficiency in workflows, a lack of connections between the fundamental business processes and the project delivery process.

Issue

Currently, many organizations utilizing ERPNext or Frappe Cloud for their core business operations face significant limitations in managing projects and quality assurance within the same ecosystem. Project tracking, task management, and testing processes are often handled using external tools such as Jira or Trello, which are not natively integrated with ERP systems. This separation creates data silos, resulting in fragmented information flow between departments. Teams experience delays in project updates, difficulty in synchronizing tasks with operational data, and a lack of unified visibility over project progress and testing outcomes.

Anticipated Outcomes

The implementation of the Project and Test Management System within ERPNext/Frappe Cloud is expected to streamline project execution, improve collaboration, and enhance quality assurance processes. Project management efficiency is projected to increase by up to 60% through the automation of task assignments, progress tracking, and reporting all within a single, integrated platform. The system will eliminate the need for external tools such as Jira, reducing operational costs, and data fragmentation. Test management capabilities will enable teams to document, execute, and monitor test cases directly in the ERP environment, leading to improved traceability and faster defect resolution. Dashboards will provide real-time visibility into project timelines, workloads, and quality metrics, supporting data-driven decision-making. Overall, these outcomes will enhance productivity, ensure better alignment between departments, and promote consistent project delivery within one unified system.

Recommendation

Develop and deploy an integrated Project and Test Management module within ERPNext/Frappe Cloud to consolidate all project-related and quality assurance activities. The module should include functionalities for project creation, task scheduling, test case documentation, issue tracking, and performance reporting. Integration with ERPNext's existing modules such as Human Resources, Accounting, and Inventory should be prioritized to ensure seamless data flow across departments. Implementation should follow a phased approach: requirements gathering, system design, module development, user testing, and deployment. Adequate training sessions and user manuals should be provided to ensure smooth adoption among project managers, developers, and QA teams. Continuous monitoring and user feedback collection are recommended for ongoing system improvement.

Justification

The introduction of an integrated Project and Test Management System directly addresses the inefficiencies caused by disconnected project tracking and testing tools. Currently, organizations using ERPNext rely on third-party applications for managing tasks and quality assurance, leading to duplicate data entries, higher costs, and delayed communication between teams. By incorporating these capabilities natively into the ERP environment, organizations can achieve complete workflow integration from planning and execution to quality validation within a single platform.

This solution aligns with the goal of optimizing organizational performance through automation and data centralization. It enhances collaboration across departments, ensures project accountability, and provides accurate, real-time insights for management decision-making. Moreover, it supports the long-term objective of creating a scalable, cost-efficient, and fully integrated enterprise ecosystem powered by Frappe Cloud.

Business Case Analysis Team

Role	Description	Name
Scrum Master (Project Manager)	Leads the team by guiding the project, removing blockers, and making sure everyone follows the Scrum process.	Daniel Balubar
Product Owner	Acts as the voice of the customer by defining project requirements, prioritizing the product backlog, and ensuring the team delivers value to the stakeholders.	Hans Edwin Maybtiuin

Scrum Team Member (UI/UX Designer)	They are responsible for designing the user interface of the project and test management system, making sure it is easy to use and visually clear.	Hans Edwin Maybtiuin, Carl Agot, Noel Catabona
Scrum Team Member (Developer)	He is responsible for developing the system features, writing the code, and ensuring all parts of the project and test management system function properly.	Carl Agot
Scrum Team Member (Quality Assurance)	He is responsible for testing the system, finding bugs, and helping to make sure the KPI's and real time data work correctly and reliably.	Noel Catabona
Stakeholder	Provides key input on project scope, requirements, and priorities. Approves the system's readiness for deployment and supports the transition from development to implementation.	Bary F. Reyes
Class Adviser	Offers guidance and support to the Project Manager and team, ensuring the project aligns with academic standards and requirements.	Prof. Jose Eugenio Quesada
Project Adviser	Provides technical and strategic advice to the team, ensuring that the project delivers value and meets the objectives agreed upon by the Product Owner and Scrum Team.	Felino Calderon

Problem Definition

Problem Statement

As organizations continue to expand their operations and manage increasingly complex projects, the limitations of existing projects and quality management processes within ERPNext/Frappe Cloud have become evident. The current system lacks integrated functionalities for project tracking, task coordination, and test management, forcing teams to rely on external tools such as Jira.

Without a platform for both project execution and testing activities, teams face challenges in maintaining transparency, accountability, and real-time visibility into project performance. Quality assurance processes are often managed manually or through disconnected applications, resulting in difficulty tracking test cases, identifying defects, and verifying project deliverables. These inefficiencies contribute to slower project completion times, increased operational costs, and reduced overall productivity. The absence of a unified Project and Test Management System within ERPNext compromises the organization's ability to achieve seamless collaboration and data-driven project oversight.

Organizational Impact

The proposed Project and Test Management System will transform how organizations utilizing ERPNext/Frappe Cloud plan, execute, and monitor their projects. By integrating project tracking and test management directly within the ERP environment, the system will centralize workflows, eliminate the need for third-party tools, and streamline collaboration between project managers, developers, and quality assurance teams. This integration will ensure that all project data from task assignments to testing outcomes are stored and managed within a single, unified platform, improving transparency and traceability across the organization.

With the adoption of this system, internal processes will become more structured, and data driven. The roles of project managers, developers, and QA personnel will shift toward strategic oversight, performance monitoring, and continuous improvement rather than repetitive administrative work. Automated reporting and real-time dashboards will reduce manual data entry and allow management to make informed decisions faster. Additionally, test management integration will enhance product quality and ensure that deliverables meet organizational standards before deployment.

The successful implementation of this system will result in a more efficient, collaborative, and scalable organizational workflow. By unifying project and quality management under one ERP platform, the organization will reduce operational costs, improve productivity, and strengthen overall project delivery performance supporting its vision of achieving excellence through innovation and system integration.

Technology Migration

The migration of the Project and Test Management System within ERPNext/Frappe Cloud will be conducted in a phased and controlled manner to ensure a smooth transition and minimal disruption to ongoing organizational operations. Since the system is a custom module built on top of ERPNext, it will leverage the existing cloud infrastructure, authentication mechanisms, and data management framework of Frappe. This will focus on setting up the ERPNext development environment in the Frappe Cloud, configuring the required Doctype's, user roles, and permissions for project managers, testers, and developers. This will include the deployment of the **Project Management Module** and **Test Management Module**, ensuring full compatibility with existing ERPNext components such as the user database, HR, and reporting modules.

Project Overview

Project Description

This project aims to address the inefficiencies and fragmentation in managing projects and quality assurance processes within organizations that rely on ERPNext/Frappe Cloud.

Currently, many businesses use separate third-party applications such as Jira to manage tasks and testing activities, redundant data entry, and limited visibility across departments.

The Project and Test Management System is designed to provide an integrated, cloud-based solution that unifies project execution and test management directly within ERPNext. The system introduces two key elements, Project Management and Test Management that enable task tracking, automated progress monitoring, and streamlined quality assurance processes. Through this path, project managers, developers, and testers can collaborate within a single platform, reducing reliance on external tools and ensuring data consistency across all operational layers.

By making the most of ERPNext's framework and user authentication capabilities, the system enhances productivity, transparency, and decision-making through real-time dashboards and key performance indicators (KPIs). The Project and Test Management System aim to refine project delivery and improve product quality.

Goals and Objectives

This project aims to improve project execution efficiency, quality assurance, and decision-making within organizations that utilize ERPNext/Frappe Cloud by developing an integrated Project and Test Management System. The primary goals of the project are as follows:

1. **Enhance Project Execution** – Streamline task assignment, progress tracking, and project monitoring to ensure timely and organized project completion.
2. **Increase Product Quality** – Integrate structured test management features to monitor test suites, test cases, and reports.

3. **Improve Decision-Making** – Provide real-time dashboards and key performance indicators (KPIs) that enable data-driven insights for project managers and stakeholders.
4. **Leverage Existing Framework** – Utilize ERPNext's/frappe built-in authentication, database, and module structure to minimize development time and ensure system stability within a secure environment.

Project Assumptions

The following are the preliminary assumptions for the project:

Access Control:

Authorized users including Project Managers, Developers, and Testers—will have secure access to the system based on predefined roles within ERPNext. Role-based permissions ensure that users can only view and modify data relevant to their duties, maintaining data integrity and accountability across the platform.

Resource Availability:

The organization will have sufficient resources, including the ERPNext server instance, cloud hosting environment, and necessary development tools, to support the system's customization, deployment, and maintenance. The project assumes the availability of administrative access to ERPNext/Frappe Cloud and consistent internet connectivity for all participating users.

User Adoption and Training:

End users including Project Managers, QA Testers, and Team Members—will actively participate in system orientation and training sessions. The success of the system depends on user engagement, understanding of workflows, and willingness to transition from external tools to the integrated ERPNext environment.

Project Constraints

Scope Limitations: This project focuses on developing and integrating a Project and Test Management System within the ERPNext/Frappe Cloud environment. It includes two core modules, Project Management for project creation, task creation, tracking, and team collaboration, and Test Management for managing test suites and test cases. The system will utilize ERPNext's existing user authentication, database management, and dashboard framework. However, the project excludes large-scale ERP customization, mobile application development, external API integrations beyond ERPNext, and advanced automation features such as predictive analytics or AI-based task scheduling. These functions may be considered for future system enhancements once the core modules are fully deployed.

Time Constraints: The project must be completed within the academic term, with each phase meeting its scheduled deadline to ensure timely submission and progress tracking.

Operational Budget: The project is developed with minimal budget allocation. Since it is hosted on Frappe Cloud, subscription costs, cloud resource limits, and system maintenance requirements must be managed within free or low-cost tiers. Additional costs related to data storage, plugin dependencies, or framework upgrades are outside the current project scope.

Resource Availability: The development team consists of one lead developer and multiple supporting members working on part-time schedules using personal computers. Hardware and internet limitations may affect development efficiency. Additionally, as the project relies on

ERPNext's open-source framework, system performance and compatibility are dependent on the stability and version of the underlying Frappe platform.

Major Project Milestones

Milestone	Target Date	Key Activities Included
Project Start	8/6/2025	Initial kickoff
Initiation	8/12/2025	Business Case, Project Charter, Stakeholder Plans
Planning	8/14/2025	Project team setup, project plan development
Analysis and Design	8/25/2025	UI Design, UX Design,
Development	09/13/2025	Front-end, Back-end development, Quality Assurance
Implementation	10/20/2025	Execution of transition, control areas (scope, cost, schedule, risk, integration testing, and deployment)
Project Completion	10/31/2025	Quality training, turnover, lessons learned, updates, file/document archiving

Strategic Alignment

This project directly supports the organization’s mission to enhance efficiency, collaboration, and quality assurance through technology-driven process integration. By developing a Project and Test Management System within the ERPNext/Frappe Cloud environment, the initiative aligns with the goal of optimizing project execution and improving overall operational performance. The system’s modules for project tracking, task management, and test case execution reinforce the organization’s commitment to structured workflows, accountability, and data transparency.

Furthermore, this project aligns with the broader vision of promoting digital transformation within enterprise operations. By leveraging ERPNext’s existing infrastructure, the system encourages sustainable and scalable growth while minimizing the reliance on costly third-party software. This initiative also embodies the principles of Sustainable Development Goal (SDG) 9 Industry, Innovation, and Infrastructure, by fostering innovation through integrated information systems and supporting organizations in building resilient, technology-enabled management processes.

Cost Benefit Analysis

Action	Action Type	Description	First Year Cost
System Development	Cost	Creating a project and test management in ERPNext/Frappe	₱ 0

Hosting and Infrastructure	Cost	Covers hosting services and depends on the subscription	₱ 1,500- 17,000
Contingency & Enhancements	Cost	Reserved for future updates, new tools, or unforeseen fixes	₱ 4,000 – 8,000

Alternative Analysis

1. Maintain Current Manual Process

- Description: Continue managing projects, tasks, and testing activities through third-party tools, maintaining separate systems for project tracking and quality assurance.
- Pros: No immediate development cost.
- Cons: Some third-party tools have limits, for example Jira has a limit of 10 members up and after that they must pay.
- Conclusion: This option partially addresses project management needs but does not support the full scope of integrated projects and test management required for modern development workflows.

2. Develop a Custom Project and Test Management System within ERPNext/Frappe Cloud

- Description: Build a fully integrated solution within ERPNext by developing custom modules for Project Management and Test Management, ensuring seamless workflow, data, and role-based access control.
- Pros: Unified platform for project execution, testing, and reporting.

- Cons: Requires moderate technical expertise for customization and testing.
- Conclusion: Provides the most efficient, scalable, and sustainable solution.

Project Charter

Project Start Date	August 08, 2025	Project End Date	October 31, 2025	
Project Title	DHAN- Project and Test Management System		Project Manager	Hans Maybituin
PROJECT INFORMATION				
This project aims to create a Project and Test Management System within ERPNext to improve visibility and collaboration.				
Project Scope			Deliverables	
Development of project tests and management systems.			Deployment of Project and Test Management System, budget setup, documentation, and training.	
Risks and Issues			Assumptions/Dependencies	
Potential delays and technical limitations.			Assumptions: Assumes no major feature additions will be added in the approved scope. Dependencies: Project depends on the stability and support of the ERPNext/Frappe environment.	
PROJECT OBJECTIVES				
Objective ID		Objective Description		
OBJ-01		Streamline task assignment, progress tracking, and project monitoring to ensure timely and organized project completion.		
OBJ-02		Integrate test management features to monitor test suites, test cases, and reports.		
OBJ-03		To use ERP Next's built-in authentication, database, and module structure to minimize development time and ensure system stability within a secure environment.		
OBJ-04		Enhance transparency, accountability, and real-time progress visibility.		
MILESTONES				

MILESTONES		Key Activities / Description	Actual Date
Project Start		Team orientation	August 8, 2025
Initiation Phase		Daily Meeting with stakeholders	August 12–15, 2025
Planning Phase		Project plan, schedule	August 25, 2025
Analysis & Design Phase		UI/UX design, and system planning	September 5, 2025
Executing Phase		Coding and testing the system	September 20, 2025
Monitoring and Controlling		Testing, Debugging	October 10, 2025
Project Completion		Final presentation	October 31, 2025
PROJECT TEAM		APPROVAL/REVIEW COMMITTEE	
Project Manager	Hans Maybituin	Client	Barry F. Reyes
Technical Lead	Carl Agot	Project Adviser	Felino Calderon
Team Members	Daniel Balubar Noel Catabona		

Project Purpose/Justification

The purpose of this project is to develop an efficient and user-friendly system that assists academic and software development teams in managing their projects and test cases digitally.

Currently, project tracking is done manually using documents, spreadsheets, and messaging apps, which makes it difficult for instructors or project managers to monitor student progress in real time. These methods lead to miscommunication, loss of data, and delays in deliverables.

By developing the Project and Test Management System, teams will have access to a platform that supports structured planning, automated tracking, and reporting. This not only enhances productivity but also ensures that projects meet quality standards.

Business Need/Case

Other organizations rely on connected systems to manage projects, but many still depend on separate tools for project management and quality assurance. This setup often leads to human errors, data mismatches, and unnecessary costs. Companies that use ERP systems like Frappe or ERPNext to handle their business data often turn to third-party tools such as Jira to manage tasks and testing. However, this approach creates a hole between planning, execution, and quality control, making it difficult to maintain a smooth workflow.

To address these challenges, the team DHAN Project and Test Management System was developed to bring everything together under one platform. Built within ERPNext, it allows teams to plan projects, assign and track tasks, manage test cases, and report issues, all in one place. This not only saves time but also ensures that every part of the project is connected, transparent, and easy to monitor. Through this solution, the team aims to:

- Streamline project execution and monitoring
- Reduce the need for expensive external tools
- Improve product quality through structured testing
- Enhance collaboration and communication between teams
- Support smarter, data-driven decision-making

DHAN provides a smarter, faster, and more connected way to manage projects and ensure quality by helping teams work efficiently and maintain exacting standards throughout the development process.

Business Objectives

- **Integrate project and test management** into a single, unified platform within ERPNext to eliminate dependency on external tools.
- **Enhance project execution efficiency** by providing real-time visibility of tasks, timelines, and team progress through a dashboard.
- **Improve product quality** by introducing structured test case management, execution tracking, and issue reporting features.
- **Reduce data fragmentation and workflow inefficiencies** caused by using separate systems for project management and quality assurance.
- **Empower data-driven decision-making** with accessible performance metrics and key indicators that guide managers and teams.
- **Using the ERPNext framework** to maximize existing infrastructure, ensuring scalability, cost-effectiveness, and seamless integration.

These objectives support the organization's goal to modernize its internal systems, promote collaboration, and deliver high-quality project outcomes through a connected and efficient platform.

Project Description

The purpose of this project is to create a Project and Test Management System within the current ERPNext platform for DHAN. The system is tasked to cover the failures resulting from separated project management tools and quality assurance processes in companies. Presently, most firms operate hybrid workflows having primary business operations on ERP systems while using external software such as Jira for project monitoring and testing. This practice leads to increased expenses and workflows.

The provided system will act as a platform for controlling project implementation and test assurance activities, which will offer a single environment for joint working by project managers, testers, and team members. With modules for Project Management and Test Management, users can plan, allocate, and track project tasks, manage test cases, run tests, and report within one interface.

Constructed in the Frappe framework above ERPNext, the platform will draw upon built-in user authentication and data management functionality to offer a reliable and scalable solution. The real-time dashboards will provide stakeholders with a transparent view of project development, test results, and key performance indicators (KPIs), facilitating data-driven decision-making and greater accountability throughout the entire project life cycle.

Project Objectives and Success Criteria

The main goal of this project is to design and create an **integrated Project and Test Management System** within the **ERPNext framework** to streamline project needs and quality assurance. The objectives follow the **SMART** format to ensure clarity, feasibility, and measurable progress.

Specific: Its aim is to eliminate fragmented workflows with the consolidation of test management as well as project management processes under one common ERPNext-driven platform. It is set to provide modules for tracking tasks, test running, as well as tracking issues with a dashboard.

Attainable: These goals are achievable using the existing ERPNext and Frappe framework, supported by structured development phases that include research, design, testing, and deployment.

Realistic: The objectives align with the organization's current challenges such as unscalable external tools and disconnected quality processes and the team's technical expertise in ERPNext customization.

Time-bound: The project is expected to be completed within **one academic year**, following a structured schedule that includes design, development, testing, and evaluation phases.

Requirements

Project and Test Management System

- Users will be able to create, assign, report, and monitor project tasks through a dedicated dashboard.
- Project Managers can manage team members, create projects, create tasks, and monitor project progress in real time.
- Testers can execute and track test cases, while reporting issues directly within the system.
- The system will automatically show all project and testing activities for the respective user/roles for transparency and traceability.

Technical Specifications

- The system will be developed within the **ERPNext framework**, using the **Frappe** backend for modular and scalable design.
- All project and testing data will be stored securely in the ERPNext database with built-in access control.
- Dashboards will provide real-time visualization of project status, test execution results, and open issues.
- The system will be fully web-based, accessible through modern browsers, and optimized for both desktop and mobile devices.

Users and Access Levels

- **Project Manager:** Full access to project creation, task management, team assignment, and performance dashboards.

- **Tester:** Access to test management modules for creating, executing, and reporting test cases and issues.
- **Team Member/Developer:** Limited access to assigned tasks, updates, and issue tracking.
- **System Administrator:** Responsible for managing user roles, permissions, and configurations within ERPNext.

Constraints

People: The project is being developed by a student team with limited members and availability.

Budget: The system is created as part of an academic requirement, so there is no allocated budget; however, potential costs such as server hosting and maintenance may arise.

Time: The entire development process must be completed within **one academic year**, with strict deadlines for each phase of research, design, development, and testing.

Equipment: The team will utilize personal laptops and available online tools, which may not match professional development environments in performance.

The team aims to deliver a functional, efficient, and high-quality system by managing resources and time effectively.

Assumptions

The following assumptions are made for the successful execution of the project:

- The client organization will provide access to necessary **projects and test-related data** for development and testing purposes.
- **Stakeholders** will be available to participate in testing, validation, and feedback sessions.
- The **ERPNext environment**, hosting platform, and internet infrastructure required for deployment will be readily available and stable.
- All users involved will have sufficient training or orientation on using the system once deployed.

Preliminary Scope Statement

The DHAN Project and Test Management System entail the planning, prototyping, and execution of a consolidated system integrating the management of projects and quality assurance within the ERPNext system. The system shall include the key features of the management of tasks, tracking of projects, preparation, and execution of test cases, reporting of defects, and real-time analytics of the dashboards.

The purpose of this project revolves around creating modules ensuring increased collaboration, transparency, and efficiency while managing projects from the planning stage to the testing stage.

It shall eliminate fragmented third-party applications and replace them with an integrated, single place where all the projects and the processes of the tests are interconnected.

However, the project scope excludes large-scale ERP customizations unrelated to project and test management, external API integrations beyond ERPNext's default modules, and mobile app development. Deployment will be carried out within the existing ERPNext instance and hosting environment, ensuring full compatibility with its existing features and user management system.

Risks

The identified future risks are the ones for the DHAN Project and Test Management System. Appropriate risk prevention and avoidance strategies shall be applied by the team and the project manager to mitigate the probability and the effect of the risks:

- **Data Inconsistencies:** Incomplete or inaccurate project and testing data may affect the reliability of dashboards, reports, and performance metrics.
- **System Security Risks:** Since the system operates within an ERPNext environment, improper configuration or access control could expose sensitive project information to unauthorized users.
- **User Adaptation Challenges:** Team members or testers unfamiliar with ERPNext or integrated management tools may initially struggle to adjust to the new system and workflows.

- **Technical Dependencies:** Unexpected compatibility issues with the ERPNext framework, or version updates may cause development delays or require additional debugging.
- **Limited Resources and Time Constraints:** Given the student-led nature of the project, restricted development time and equipment limitations could affect feature completion or testing quality.

Project Deliverables

The following deliverables must be completed upon the successful implementation of the **DHAN Project and Test Management System**. Any modifications or additions to these deliverables must be reviewed and approved by the project sponsor or adviser.

- Design and Development of the Project and Test Management System
 - Project Management Module
 - Test Management Module
 - Role-Based Access Controls
 - Comprehensive Documentation

Summary Milestone Schedule

Milestone Phase	Key Activities	Target Date
Project Initiation	Official commencement of the project is complete.	Aug 8, 2025
Planning Phase	• Project planning activities have begun. • Finalization and approval of the overall project plan.	Aug 16, 2025
Design Phase	• Planning of user interface theme and assignment of user roles. • Completion of UI/UX designs for development.	Sept 2, 2025
Development Phase	• Release of initial working version (prototype). • Development of final, improved prototype based on client preference.	Sept 19, 2025
Deployment Phase	Full rollout and implementation of the system in the live environment.	October 30, 2025

Closure	Project concludes with handover, documentation, and formal wrap-up activities.	October 30, 2025
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Summary Budget

The table below outlines the key cost elements and their estimated amounts for the project.

Most activities incur no charges except for hosting and infrastructure expenses related to service subscriptions.

Summary Budget – List component project costs

Project Component	Component Cost
Development & Setup	₱ 0
Data Migration	₱ 0
Hosting & Maintenance	₱ 7, 000
Testing & Enhancements	₱ 0
Total	₱ 7, 000

Project Approval Requirements

The project will be considered successfully completed once the DHAN Project and Test Management System has been fully developed, tested, and deployed within the ERPNext environment. Approval will be based on verified evidence of the system's functionality, reliability, performance, and user satisfaction during the evaluation phase.

Project Manager

The DHAN software development team Project Manager will be **Hans Edwin Maybituin**. He oversees the entire project of coordination, schedule, communication, task distribution, and tracking progress.

The Project Manager shall ensure all the work of development is as per the technical requirements of the ERPNext system and the academic needs of the course.

Stakeholder Management Strategy

Introduction

The Stakeholder management of creating the DHAN Project and Test Management System focuses on the identification, involvement, and management of all stakeholders effectively throughout the whole duration of the project. It promotes open communication, transparency, and teamwork to make sure the desired goals of the project, outputs, and results are aligned both organizationally and academically.

Objectives

This strategy aims to:

- Identify and prioritize all relevant stakeholders.
- Understand stakeholder roles and expectations.
- Maintain open and transparent communication.
- Provide proper feedback and conflict resolution channels.
- Build continuous stakeholder trust and cooperation.
- Include stakeholder feedback.

Identify Stakeholders

DHAN Project and Test Management System stakeholders are individuals and groups involved in, affected by, or able to impact the system's build, design, and delivery. These stakeholders are all responsible for ensuring that the project complies with both academic needs and organizational needs.

Stakeholder identification process:

- Conduct internal discussions with team members and Project Leaders.
- Review current project management and testing challenges.
- Hold consultation sessions with the project adviser, clients, and end users.

Stakeholders Categories

Internal Stakeholders: Project Manager, Developers, Testers, and Team Members.

External Stakeholders: Project Adviser, Academic Evaluators, and End Users.

Key Stakeholders

Project Manager and Development Team (Internal): Responsible for the design, development, and testing of the DHAN Project and Test Management System.

Testers and Team Members (Internal): Serve as the main users of the system during testing and evaluation.

Project Adviser (External): Provides academic oversight, guidance, and validation of the project's progress and outcomes to ensure alignment with course objectives.

Academic Evaluators (External): Review and assess the final system to verify that it meets both technical and academic standards.

End Users / Organizations (External): Represent future users or institutions that may adopt the ERPNext-based Project and Test Management System for real-world applications.

Stakeholder Analysis

Stakeholder	Power / Interest Level	Engagement Strategy
Prof. Bary F. Reyes	High Power / High Interest	Maintain regular communication and progress updates through meetings and demonstrations. Seek feedback during major milestones to ensure alignment with expectations.
Development Team	Medium Power / Medium Interest	Conduct weekly team meetings, task reviews, and sprint planning sessions to ensure collaboration, accountability, and continuous progress.
Project Adviser	Medium Power / Medium Interest	Provide regular updates on development progress and testing outcomes. Request guidance and validation during each phase to align with academic standards.
End Users	Low Power / High Interest	Conduct user testing sessions, collect feedback on usability and functionality, and make necessary adjustments before deployment.

Power Interest/Grid

High Power / High Interest (Manage Closely): Prof. Bary F. Reyes, Development Team
High Power / Low Interest (Keep Satisfied): Project Adviser
Low Power / High Interest (Keep Informed): End Users

Stakeholder Engagement Plan

Stakeholder	Engagement Method	Frequency	Purpose
Project Manager and Development Team	Team meetings, progress reports, and task tracking via ERPNext.	Twice weekly	Coordinate development tasks, monitor progress, and address technical issues.
Testers and Team Members	Testing sessions, feedback discussions.	Weekly	Evaluate usability and functionality, report issues, and suggest improvements.
Project Adviser	Consultations, progress reports, and feedback meetings.	Monthly or as needed	Provide academic validation, ensure alignment with project goals, and offer technical guidance.
Academic Evaluators	Evaluation reports, final presentations, and documentation review.	End of each project phase	Assess project quality, functionality, and academic compliance.
End-users Organizations	Demonstrations, feedback surveys, and post-deployment reviews.	After major milestones	Gather feedback on usability, performance, and system effectiveness for real-world applications.

Communication Plan

Communication Type	Audience	Format	Owner	Frequency
Kick-off Meeting	All Stakeholders	Virtual / In-person	Project Manager	At the start of the project
Sprint Review Meetings	Project Adviser, Development Team	Presentation / Discussion	Project Manager & Dev Team	End of each sprint
Progress Reports	Project Adviser, Academic Evaluators	Digital Reports / Documents	Project Manager	Bi-weekly
Testing Feedback Reports	Testers, Development Team	Digital Forms / Reports	QA Lead or Assigned Tester	After each testing cycle

Training and Demonstration Sessions	End Users / Organizations	Interactive / Recorded Sessions	Tech Lead / Developer	Pre-deployment
Final Presentation and Submission	Adviser, Academic Panel	Presentation and Documentation	Project Manager & Team	End of project phase

Risk Mitigation and Response

Risk	Stakeholder Affected	Mitigation Strategy
Resistance to Change	Project Team Members, Testers	Involve users early in the development phase, provide orientation and hands-on training sessions, and conduct pilot testing to build familiarity and confidence.
Limited User Adoption	End Users / Organizations	Design a simple and intuitive user interface, provide user guides and training materials, and maintain open communication channels for feedback and support.
Data Privacy and Security Concerns	All Stakeholders	Implement strict role-based access controls within ERPNext, ensure secure handling of project and test data, and perform regular security checks and audits.
Technical Compatibility Issues	Development Team	Conduct frequent testing within the ERPNext environment, document all configurations, and maintain backups before major framework updates.
Time and Resource Constraints	Project Team	Use agile planning, prioritize essential features, and manage workload distribution to ensure timely completion of deliverables.

Scope Analysis

Introduction

This document provides the scope of the DHAN Project and Test Management System, an ERPNext-based application to be used for improving the execution of projects and quality assurance through centralization. It identifies the role of the project team, the major deliverables, and the system limits, so all the aims are always manageable and aligned both academically and organizationally.

The document follows a structured five-step approach that includes:

1. Gathering system requirements.
2. Defining the overall project scope.
3. Creating a Work Breakdown Structure (WBS).
4. Verify the defined scope.
5. Maintaining scope of control throughout development.

These steps serve as a guide for the development team to ensure that the system is delivered on time, within scope, and according to the intended objectives specifically to improve project transparency, team collaboration, and test management efficiency within the ERPNext environment.

Scope Management Approach

- **Authority and Responsibility for Scope Management**

Scope management of the DHAN Project and Test Management System falls under the responsibility of Project Manager Hans Edwin Maybituin, therefore all the tasks, activities, and outputs of the project are all within the boundaries of the given scope.

Project Advisor, Mr. Felino, provides academic guidance, orientation, and higher-order support to ensure the project achieves its academic and technological goals.

- **Scope Definition**

Project scope is outlined by primary planning documents like the Stakeholder Management Plan, specifically define the project goals, major deliverables, constraints, and limitations to retain focus throughout the software life cycle.

- **Scope Measurement and Verification**

To make sure the scope does not exceed the specified procedures:

- **Existing Documentation**

Documentation sources such as the SRS and the stakeholder requirements should be used to validate whether all system functions and tasks performed are aligned with the agreed requirements.

- **QA Reviews**

System reviews should always be performed by staff and the Project Manager so all the modules of Project Management, Test Management, and Dashboards to achieve the desired functionality, quality, and performance characteristics.

- **Project Sponsor Approval**

The Project Adviser's and Client evaluation and feedback will confirm whether all deliverables align with the intended project scope or if any deviations require corrective action.

- **Scope Change Process**

It falls upon the choice of the DHAN Development Team to recommend the project's scope for any adjustments it may require. All proposed adjustments are required to be presented and

approved by the Project Adviser, Mr. Felino, and the team since all modifications should be possible, justified, and for the purpose of the project. Any approved scope of changes shall be recorded and reported to affected stakeholders to be open and traceable.

- **Acceptance of Final Project Deliverable**

End products of the Test Management System and DHAN Project should be reviewed, signed off, and formally approved by the Project Adviser and Academic Assessors. The Project Manager should ensure all the outputs comply with the documented scope, and all the stakeholders be informed of any approved changes before final submission and presentation.

Roles and Responsibilities

Roles	Responsibilities
Project Manager	Responsible for leading the team, managing the project schedule, facilitating meetings, and ensuring the team follows agile practices.
Product Owner	Provides key input on project scope, requirements, and priorities. Approves the system's readiness for deployment and supports the transition from development to implementation.
Stakeholder	Responsible for giving important input, setting overall goals, approving the system's progress, and supporting its implementation.
Team Members	Responsible for designing, developing, and testing the system. Each member focuses on their role in helping build a functional and high-quality product.
Class Adviser	Responsible for giving academic guidance, making sure the project meets school requirements and educational objectives.
Project Adviser	Responsible for offering technical and strategic advice to ensure the billing system with chatbot meets its goals and adds real value.

Scope Definition

As signed off by the Project Adviser and in line with the project-based learning (PBL) process, the scope as defined for the DHAN Project and Test Management System is outlined below:

- The project consists of designing, developing, and deploying an ERPNext-based Project and Test Management System to enhance project execution, collaboration, and quality assurance under a platform. The system will incorporate necessary project management and testing features utilizing the ERPNext and Frappe platforms. The major features and functions are:
- Enables team members and project managers to create, delegate, and track tasks, control timelines, and monitor progress through a collaborative dashboard.
- Facilitates testers to plan test cases, run test runs, and report defects offering a formal process for quality control and defect tracking.
- Shows real-time analytics, project performance indicators, and test results to support decision-making and project assessment.
- Guarantees users (Project Manager, Tester, and Team Member) have safe access to system modules and data.
- The system will be hosted in the ERPNext environment and will adhere to Agile methodology, encouraging iterative development, ongoing feedback, and flexibility during the project life cycle.

The project scope can be adjusted as appropriate, with the prior approval of the Project Adviser and the development team, so that any changes still match the goals and requirements of the project and the stakeholders.

Project Scope Statement

The project entails the development, design, and implementation of the DHAN Project and Test Management System, a comprehensive solution developed using the ERPNext platform. The system was created to consolidate project management and quality control procedures through task tracking, testing, and reporting within a single, unified platform.

Key features include:

- **Project Management Module** for creating, assigning, and tracking tasks and timelines.
- **Test Management Module** for managing test cases, executing tests, and reporting issues.
- **Dashboard** that provides real-time visibility into project progress and quality metrics.
- **Role-Based Access Control** ensuring secure and system access for all users.
- **Data and Security Measures** embedded within the ERPNext environment.

Project Deliverables

1. **ERPNext-based Project and Test Management System.**
2. **Project Management Module** for efficient tasks and progress tracking.
3. **Test Management Module** for structured test case execution and issue reporting.
4. **Dashboard** provides real-time analytics and project insights.
5. **Role-based permissions** for project managers, testers, and team members.
6. **Project documentation**, including technical, testing, and user manuals.

Project Exclusions

The following items are outside the scope of the DHAN Project and Test Management System:

- A. Full development of unrelated ERPNext modules or features not tied to project and test management.
- B. Integration of external APIs or third-party tools beyond the ERPNext and Frappe framework.
- C. Implementation of advanced analytics or machine learning components outside the current system goals.
- D. Development of a standalone mobile application version.

Project Constraints

The following constraints may affect the project's implementation and must be considered throughout development:

- The project must be completed within one academic year, with each development phase of research, design, implementation, testing, and evaluation meeting its scheduled deadlines.
- The project is developed by a DHAN team with limited time and availability, requiring efficient task scheduling and collaboration.
- The project is created as part of an academic requirement and with a small budget. However, potential future costs such as hosting, server maintenance, or deployment infrastructure may arise.
- The development team uses personal laptops and freely available tools, which may limit processing performance compared to professional-grade systems.

Project Assumptions

- Stakeholders, including the project adviser and evaluators, will provide timely feedback and approvals.
- The necessary infrastructure and tools will remain available throughout the project.
- Team members will remain actively involved and committed for the full duration of the project.
- The ERPNext and Frappe frameworks will remain stable and compatible with the custom modules developed by the team.

Scope Verification

The Project Manager and the team will continuously check and confirm all deliverables to ensure they are in accordance with the sanctioned project scope and laid-down requirements. After internal validation of the deliverables, the Project Adviser will check and officially approve them through documented affirmation or evaluation reports. This helps ensure that each achieved component works constructively towards the entire project goals.

Scope Control

Scope changes will be controlled intensively using supporting documents like the Project Charter, Change Management Plan, and SRS.

If adjustments are required, the team will:

1. Evaluate the reason and impact of the proposed change.
2. Discuss it with the **Project Adviser** and development team.
3. Obtain formal approval before implementation.
4. Document all changes to maintain transparency and accountability.

This process ensures that all changes are in line with project objectives and meet academic and technical standards.

Schedule Management

Introduction

This Schedule Management Plan provides an organized methodology for controlling the development schedule of the DHAN Project and Test Management System. The objective is to ensure that all project milestones from research to deployment happen on time and according to project objectives. The project plan outlines the approach, tools, and roles for scheduling, tracking progress, and managing changes to ensure timely completion of each project phase.

Schedule Management Approach

Scheduling Tool

To plan tasks, monitor progress, and manage time efficiently, the team will make use of project management tools that align with ERPNext workflows:

Jira: Used for sprint planning, backlog management and refinement, task tracking, and visualization using burndown charts.

OpenProject: Used for Gantt chart planning, milestone scheduling, budgeting, and cross-project timeline visualization.

Both will assist the team in planning effectively, tracking work development, and tweaking as required to ensure the project remains in line with its timeline.

Schedule Milestones

Milestone	Target Date	Key Activities Included
Project Start	August 8, 2025	Official project kickoff and team orientation.
Initiation	August 15, 2025	Preparation of the project charter, defining objectives, and identifying stakeholders.
Planning	August 25, 2025	Development of project plans, task distribution, and setting timelines for each phase.
Analysis and Design	September 5, 2025	System requirement gathering, database design, and creation of the system's user interface layout.

Development	September 20, 2025	Coding of system modules (frontend and backend), integration, and initial internal testing.
Implementation	October 10, 2025	Conducting functional and user testing, addressing bugs, and deploying the system for client review.
Project Completion	October 31, 2025	Final presentation and turnover to the client, documentation handover, and project closure.

Schedule Control

To control the schedule, weekly team meetings are conducted to check progress. Team members update a shared project tracker. If any delay is expected, it must be reported immediately to allow proper adjustment. The project supervisor monitors if all deliverables are completed on time.

- **Progress Reviews:** Weekly.
- **Update Frequency:** Every week or upon task completion.
- **Communication:** Through group chat, email, and Teams meetings.
- **Responsibility:** Each team member is accountable for their assigned tasks.

Roles and Responsibilities

- **Project Leader:** Oversees schedule progress.
- **UI/UX Designer:** Designs the user interface based on the prototype.
- **Developers:** Work on both the billing system and AI features.
- **QA Tester:** Ensures features are functional and bug-free before moving to the next phase.

Schedule Changes and Thresholds

A schedule change is required if any task is delayed. For example, if a task was planned for 10 days but it might take more than 11 days, a change request must be submitted and approved. Major changes to milestones or deliverables must be approved by the project client and project adviser.

Scope Change

If new features or major changes are added to the Project and Test Management System, the schedule may need to be adjusted. The team will review how much extra time is needed and whether it affects the current progress. Any major change will need the project manager and sponsor approval to rebalance the schedule.

Cost Management

Introduction

This Cost Management Plan outlines how costs for the project will be planned, estimated, and managed throughout its lifecycle. The purpose of this plan is to ensure project expenditures are controlled and aligned with available resources.

It defines:

- Who is responsible for managing project costs
- Authority levels for approving budget changes
- Procedures for measuring and reporting cost performance
- Methods for addressing cost variances

Cost Management Approach

Cost tracking will be done per key deliverable (hosting/cloud, subscriptions, integration testing), as identified in the project's Work Breakdown Structure (WBS). Resource costs will be monitored across the development, testing, and deployment phases.

Since this is a student-led project under the Asia Pacific College, development costs are covered by academic provisions, and only operational expenses (such as hosting, and utilities) will be incurred by the client.

Measuring Project Costs

Cost performance will be measured against a baseline and evaluated monthly during check-ins or milestone reviews.

Metrics include:

- Actual Cost (AC)
- Planned Value (PV)
- Earned Value (EV)
- Cost Performance Index (CPI)

Reporting Format

The project team will submit a monthly report that includes:

- Current expenditure vs. budget baseline
- Cost performance index (CPI)
- Corrective actions (if needed)
- Updates on approved budget changes

Cost Variant Response Process

Control Threshold:

If the CPI or SPI falls below 0.8 or rises above 1.2, a **Cost Variance Corrective Action Plan** will be initiated.

Process:

- Within 5 business days, the Project Manager drafts a corrective action report with options.
- The Project Sponsor selects an option within 3 days.

- A finalized action plan is incorporated into the project scope and tracked moving forward.

Cost Change Control Process

All changes that affect the budget must follow a **formal change request** procedure, submitted to the Project Sponsor. The request must include:

- Rationale for the change
- Impact analysis
- Recommended adjustments

No budget changes will be made without written approval from the Project Sponsor.

Project Budget Estimate

Manpower Estimated Cost

Role	Monthly Salary	Project Duration	Count	Total
Project Manager (Intern)	₱20,000.00	6 months	1	₱120,000.00
UI/UX Designer (Intern)	₱15,000.00	6 months	1	₱90,000.00
Software Developer (Junior)	₱26,000	6 months	2	₱156,000.00
QA Tester (Intern)	₱12,000.00	6 months	1	₱72,000.00
Total				₱498,000.00

Maintenance and Hosting

Description	Cost (per year)	Period	Total
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Cloud Hosting (ERPNext/Frappe Cloud)	₱7,200.00	1 Year	₱6,000.00
Domain & SSL	₱3,000.00	1 Year	₱3,000.00
Backup & Security Services	₱4,000.00	1 year	₱4,000.00
Total			₱13,000.00

ESTIMATED PROJECT COST

Category	Total Cost
Manpower Estimated Cost	₱498,000.00
Hosting	₱13,000.00
TOTAL	₱511,000.00

Work Breakdown Structure

Introduction

The Work Breakdown Structure (WBS) presented here outlines all the work required to complete the Integrated System Development project. It is structured in several views for comprehensive understanding, including Outline, Hierarchical, Tabular, and Tree Structure formats. A WBS Dictionary is also included for further clarity of each element.

Outline View

1. Initiation
 - 1.1 Business Case
 - 1.2 Project Charter
 - 1.3 Stakeholders Management Strategy Plan
 - 1.4 Stakeholder Analysis
2. Planning
 - 2.1 Determine Project Team

2.2 Develop Project Plan

3. Execution

3.1 UI Design

3.2 UX Design

3.3 Tech Stack Setup

3.4 Front-end Development

3.5 Back-end Development

3.6 Quality Assurance

3.7 Implementation/Transition

4. Control

4.1 Scope Management

4.2 Schedule Management Plan

4.3 Cost Management

4.4 Resource Management

4.5 Quality Management

4.6 Risk Management

4.7 Communication Management

4.8 Change Management

5. Closeout

5.1 Project Turnover

5.2 Document Lesson Learned

5.3 Update Files/Records

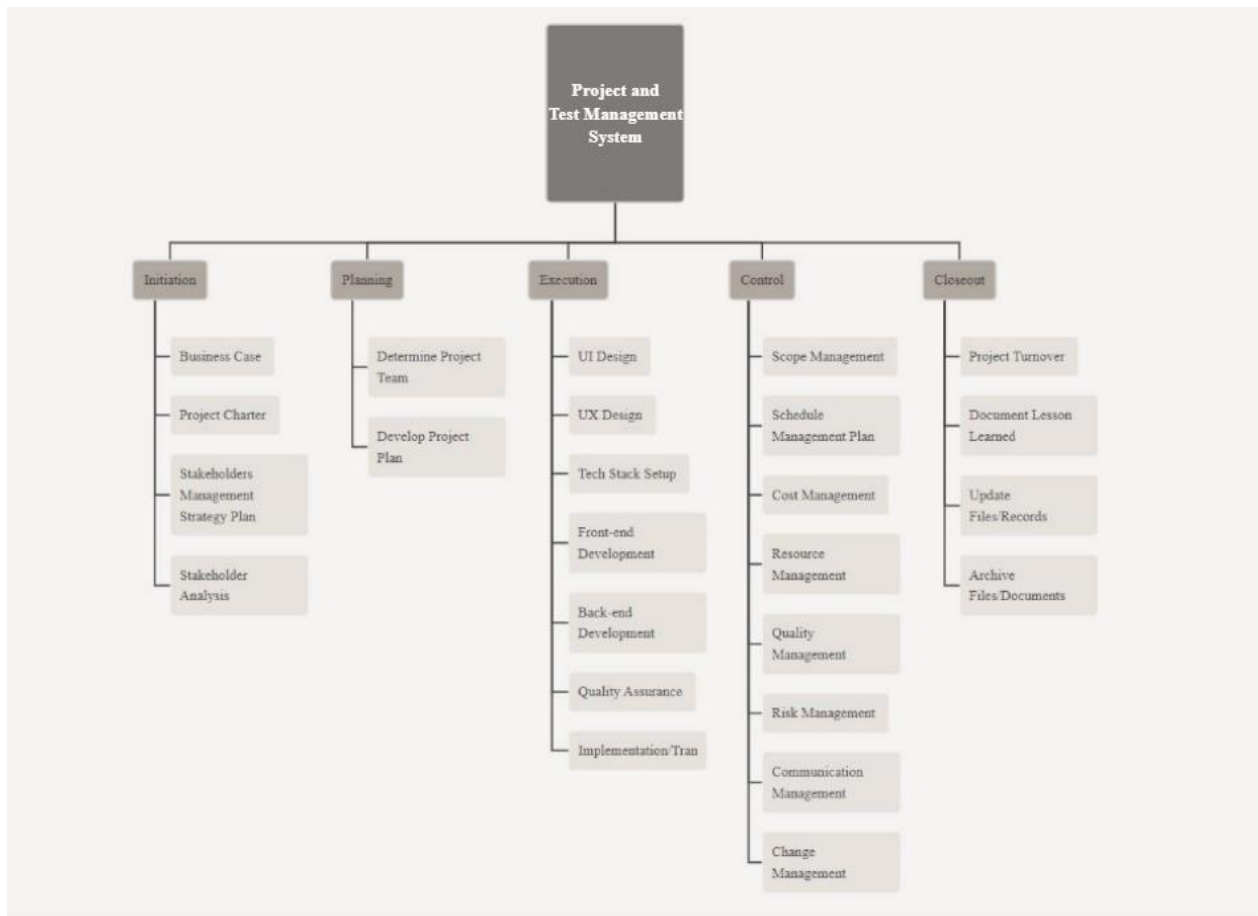
5.4 Archive Files/Documents

Tabular Structure

WBS Code	Task Name	Assigned To	Status	Priority
1	Initiation	Carl Agot	New	Normal
1.1	Business Case	Carl Agot	New	Normal
1.2	Project Charter	Carl Agot	New	Normal
1.3	Stakeholders Management Strategy Plan	Carl Agot	New	Normal
1.4	Stakeholder Analysis	Carl Agot	New	Normal
2	Planning	Carl Agot	New	Normal
2.1	Determine Project Team	Carl Agot	New	Normal

2.2	Develop Project Plan	Carl Agot	New	Normal
3	Execution	Hans Edwin Maybituin	New	Normal
3.1	UI Design	Hans Edwin Maybituin	New	Normal
3.2	UX Design	Hans Edwin Maybituin	New	Normal
3.3	Tech Stack Setup	Carl Agot	New	Normal
3.4	Front-end Development	Hans Edwin Maybituin	New	Normal
3.5	Back-end Development	Carl Agot	New	Normal
3.6	Quality Assurance	Noel Catabona	New	Normal
3.7	Implementation/Transition	Noel Catabona	New	Normal
4	Control	Daniel Balubar	New	Normal
4.1	Scope Management	Daniel Balubar	New	Normal
4.2	Schedule Management Plan	Daniel Balubar	New	Normal
4.3	Cost Management	Daniel Balubar	New	Normal
4.4	Resource Management	Carl Agot	New	Normal
4.5	Quality Management	Noel Catabona	New	Normal
4.6	Risk Management	Hans Edwin Maybituin	New	Normal
4.7	Communication Management	Hans Edwin Maybituin	New	Normal
4.8	Change Management	Noel Catabona	New	Normal
5	Closeout	Noel Catabona	New	Normal
5.1	Project Turnover	Noel Catabona	New	Normal
5.2	Document Lesson Learned	Noel Catabona	New	Normal
5.3	Update Files/Records	Noel Catabona	New	Normal
5.4	Archive Files/Documents	Carl Agot	New	Normal

Tree Structure



WBS Dictionary

WBS Code	Task Name	Description	Responsible
1.1	Business Case	Define the business needs and justify the project.	Carl Agot
1.2	Project Charter	Develop the project charter for stakeholder approval.	Carl Agot
1.3	Stakeholders Management Strategy Plan	Identify and plan stakeholder engagement.	Hans Edwin Maybituin
2.1	Determine Project Team	Assign project team roles and responsibilities.	Hans Edwin Maybituin
3.4	Front-end Development	Code and build the front end of the application.	Noel Catabona
3.7	Implementation/Transition	Deploy the system and transition to an operational environment.	Daniel Balubar
4.5	Quality Management	Establish and implement quality standards and processes.	Noel Catabona

5.2	Document Lesson Learned	Record and review lessons learned for future projects.	Daniel Balubar
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Work Packages Based on WBS

ID	Work Package	Type	Status	Assignee	Priority
6282	1. Initiation	Phase	New		Normal
6287	1.1 Business Case	Work Package	New		Normal
6288	1.2 Project Charter	Work Package	New		Normal
6289	1.3 Stakeholders Management Strategy	Work Package	New		Normal
6290	1.4 Stakeholder Analysis	Work Package	New		Normal
6283	2. Planning	Phase	New		Normal
6291	2.1 Determine Project Team	Work Package	New		Normal
6292	2.2 Develop Project Plan	Work Package	New		Normal
6294	3. Execution	Phase	New		Normal
6293	3.1 UI Design	Work Package	New		Normal
6294	3.3 UX Design	Work Package	New		Normal
6295	3.3 Tech Stack Setup	Work Package	New		Normal
6296	3.4 Front-end Development	Work Package	New		Normal
6297	3.5 Back-end Development	Work Package	New		Normal
6298	3.6 Quality Assurance	Work Package	New		Normal
6299	3.7 Implementation/Transition	Work Package	New		Normal
6285	4. Control	Phase	New		Normal
6300	4.1 Scope Management	Work Package	New		Normal
6301	4.2 Schedule Management Plan	Work Package	New		Normal
6302	4.3 Cost Management	Work Package	New		Normal
6303	4.4 Resource Management	Work Package	New		Normal
6304	4.5 Quality Management	Work Package	New		Normal
6305	4.6 Risk Management	Work Package	New		Normal
6306	4.7 Communication Management	Work Package	New		Normal

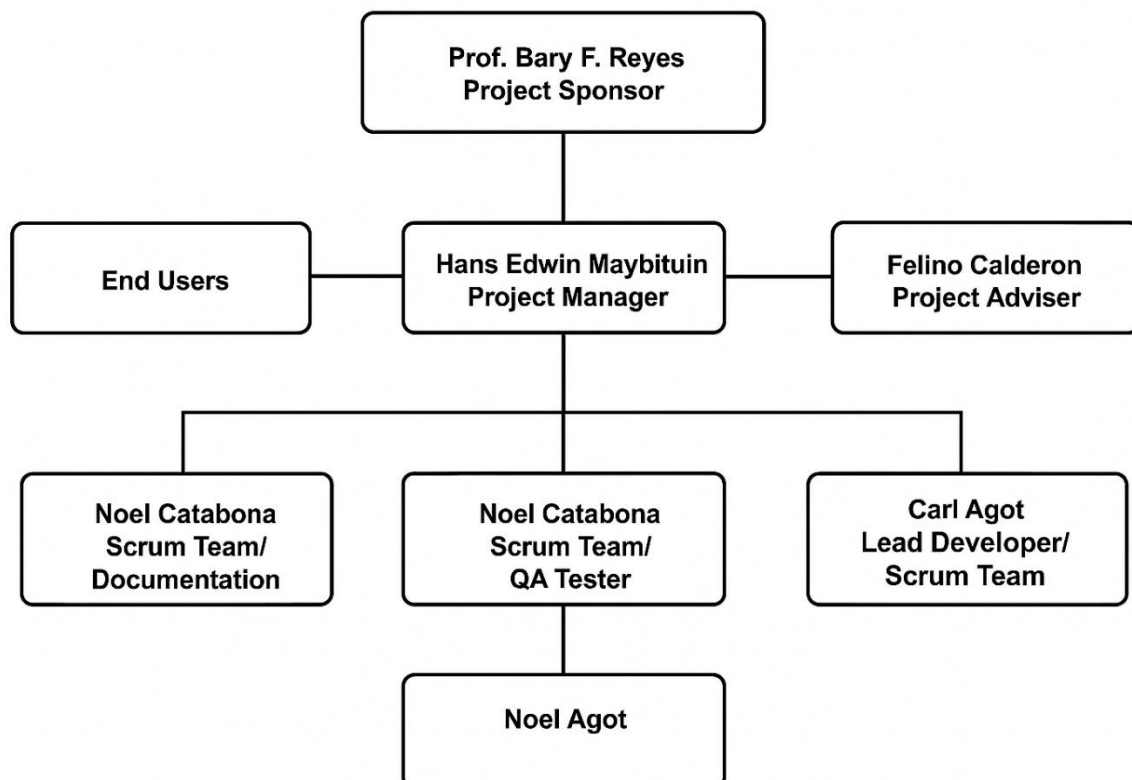
6307	4.8 Change Management	Work Package	New		Normal
6286	Closeout	Phase	New		Normal
6308	5.1 Project Turnover	Work Package	New		Normal
6309	5.2 Document Lesson Learned	Work Package	New		Normal
6310	5.3 Update Files/Records	Work Package	New		Normal
6311	5.4 Archive Files/Documents	Work Package	New		Normal

Human Resource Plan

Introduction

This Human Resource Management Plan is tailored for the development of the Project and Test Management System within the ERPNext/Frappe Cloud environment. The document outlines the strategic approach to managing human resources throughout the system's design, development, testing, and deployment phases. The project focuses on integration of project tracking and quality assurance features into ERPNext, addressing its current lack of built-in tools

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Roles and Responsibilities

Role	Authority	Responsibility	Competency
Project Sponsor	Key decision-making authority for the project; approves or denies changes in the project scope	Provides necessary resources for the project; reviews deliverables and changes throughout the project lifecycle	Should understand ISP operations, business processes, and possess strategic decision-making skills
Project Manager	Oversees the project execution	Manages the entire project lifecycle, from planning to deployment; allocates tasks and monitor's progress; facilitates communication among stakeholders	Should have advanced project management skills, technical knowledge in system development, and leadership ability
Team Members	Designs, develops, and tests the system	Works collaboratively to implement core features of the billing system and AI chatbot; participates in sprints and delivers increments of the product	Should have solid knowledge of system analysis, software development, chatbot integration, and Agile methodologies
Project Adviser	Provides support, recommendations, and guidance to the team throughout the project cycle	Guides the team in project execution, methodology selection, and implementation strategies; reviews progress and deliverables	Should be a bona fide faculty member of the School of Computing and Information Technologies at Asia Pacific College, with project advisory experience
External Users (Subscribers)	End users of the system	Participate in testing, provide feedback, and report system usability issues to improve user experience	Should be capable of navigating digital systems and communicating feedback clearly

Project Organizational Charts

The following diagram illustrates the organizational structure of the Project and Test Management System for ERPNext/Frappe Cloud.

It outlines the key roles and reporting relationships among the project team members, internal stakeholders, and external collaborators to ensure effective communication, efficient coordination, and accountability throughout the project lifecycle.

The structure highlights the integration of both project management and quality assurance functions, emphasizing collaboration across development, testing, and design teams to ensure a unified and high-quality system delivery.

Staffing Management

The project team for the Project and Test Management System consists of students from Asia Pacific College's DHAN program, working collaboratively under the project-based learning framework. Team members, collectively known as Team DHAN, serve as the primary development and testing team responsible for the system's design, implementation, and evaluation.

All roles have been distributed based on team members' skill sets to ensure balanced workloads and efficient execution. The responsibilities are defined as follows:

Role	Project Responsibility	Skills Required	Number of Staff	Performance Reviews	Recognition and Rewards
Project Sponsor	Provides necessary resources; reviews deliverables and project scope changes	Knowledge of ISP operations, business processes, and strategic decision-making skills	1	Monthly or as needed	Bonus, stakeholder recognition
Project Manager	Manages the entire project lifecycle; assigns tasks; oversees communication and progress	Advanced project management, leadership, communication, and system development knowledge	1	Monthly	Bonus or promotion
Scrum Master	Facilitates Scrum ceremonies; ensures team follows Agile principles	Agile/Scrum experience, organizational, facilitation, and communication skills	1	Weekly	Bonus or salary increase
Scrum Team	Designs, develops, and tests the billing system and AI chatbot	System analysis, software development, chatbot integration, collaboration, and Agile knowledge	Varies depending on project scope	Weekly	Bonus, salary increase, or academic recognition
Documentation Team	Prepares and maintains project documentation including manuals and system design references	Technical writing, knowledge of documentation standards, and system process understanding	1–2 (depending on project size)	Bi-weekly or as needed	Certificates, bonus, or academic recognition

Project Adviser	Guides methodology selection, implementation, and reviews outputs	Project advisory experience, analytical skills, and technical expertise	1 (faculty member)	As needed	Letter of commendation, academic recognition
External Users	Participates in testing and provides feedback to improve system usability	Ability to navigate systems and communicate feedback clearly	Open (volunteer or selected subscribers)	During user testing phase	Public acknowledgement or access benefits

Quality Management

Introduction

The team adopted an Agile development methodology throughout the creation of the Project and Test Management System for ERPNext/Frappe Cloud. The system was developed incrementally, with regular testing and feedback after each sprint. This iterative process allowed the team to adapt quickly and make continuous improvements based on insights from Prof. Bary F. Reyes, who serves as the project client and advisor.

Quality Management Approach

The team employed an Agile methodology throughout the development process. This involved building the system incrementally and conducting tests at each stage. This approach facilitates continuous improvement and allows for timely adjustments based on feedback from the client or academic advisor.

Below is an overview of team responsibilities:

Role	What They Do
Project Manager	Oversee project planning, scheduling, and coordination to make sure goals are achieved within the set timeline.
UI/UX Designer	Designs the interface and ensures that the system layout is clear, user-friendly, and consistent with ERPNext's framework.
QA Tester	Test distinct parts of the system, identify bugs, and verify if fixes are properly implemented.
Developers	Build and integrate the project and test management modules, ensuring functionality and fixing issues found during QA.
Product Owner	Reviews development progress, approves completed features, and communicates updates to Prof. Bary F. Reyes.
Client (Prof. Bary F. Reyes)	Gives feedback and helps the team know what they need from the system.

Quality Requirement and Standards

To make sure the system is high quality, the team followed these standards:

- System pages must be loaded in under 3 seconds.
- The application should maintain 99.9% uptime, with daily backup procedures.
- The interface must function smoothly across Chrome, Firefox, Safari, and Edge.
- All forms, dashboards, and reports should display accurate and updated data.
- Code should be well-documented, readable, and maintainable for future system enhancements.

Quality Assurance (QA)

To make sure the project is working well:

- The team reviewed the system requirements to check if each feature was added correctly.
- The team did several types of testing:
- Unit testing for small parts of the system.
- System testing to check everything that works together.

The team created test cases for each function of the system and checked and ran the testcases to see the result of the QA of the system if it passed or not.

Quality Control

To maintain consistency and reliability, the team implemented the following practices:

- **Code Reviews** – Developers reviewed each other's work to identify errors early and maintain quality standards.

- **Bug Tracking** – Issues were logged and monitored through ERPNext's Issue Management module.
- **Client Demonstrations** – Regular sprint reviews were conducted with Prof. Reyes to highlight progress and gather feedback.
- **Weekly Meetings** – The team discussed development progress, testing results, and areas for improvement.

Quality Control Measurements

The team tracked the following key indicators to assess the system's quality:

What The group Checked	Goal
Project Dashboard	The right information/data is shown on the landing page or the dashboard. Users can edit with the right access.
Test Dashboard	The right information/data is shown on the landing page or the dashboard. Users can edit with the right access.
Uptime	The site is up and running.
Client Satisfaction	The client is satisfied with the results of the look and functionality.
Testing Coverage	All the testcases passed for our system, making them ready for deployment.

By adhering to this plan, the team ensured that the project was not only completed successfully but also met standards of reliability, and user-friendliness.

Communication Management

Introduction

This plan aims to ensure that all communications are clear, timely, and aligned with stakeholder expectations. It outlines the communication process, including the methods of dissemination, frequency, responsible team members, and the platforms to be used. It also establishes expectations for stakeholder engagement, data confidentiality, and the allocation of communication resources.

Specifically, this plan defines:

- The type of information to be shared and the processes for dissemination
- Including platforms (e.g., Teams, Messenger).
- Along with the assigned point of contact for each communication
- Stakeholder-specific communication requirements
- Standardized communication templates and documentation formats
- A defined escalation process for resolving issues when communication fails

Following this plan, Team ensures that all parties are well-informed, engaged, and able to contribute effectively to the project's success.

Communication Management Approach

The DHAN project team primarily uses Microsoft Teams for real-time communication, conducting daily scrum meetings and group chats that help keep everyone aligned on progress, obstacles, and upcoming tasks, while fostering transparency and quick issue resolution. To track project progress and task assignments, Jira is utilized, allowing members to update task statuses,

report issues, and monitor deadlines, providing clear visibility into overall project health. For project planning, scheduling, and workflow collaboration, the team relies on OpenProject, which helps organize tasks, dependencies, and milestones to maintain a structured delivery process. Additionally, for quick clarifications and urgent questions outside formal meetings, the team uses Facebook Messenger for group chats, providing a convenient and accessible platform for fast communication.

Communication with the client is managed by Carl Agot, who serves as the main point of contact. The team conducts regular Zoom meetings to provide project updates, review progress, gather feedback, and address any concerns. This ensures consistent collaboration and alignment with client expectations throughout the project.

Communication Management Constraints

The following are key constraints that may impact the effectiveness and consistency of communication throughout the project lifecycle:

1. **Schedule Constraints:** Team members balance academic responsibilities alongside the project, which limits availability for synchronous communication and requires careful scheduling of meetings and deadlines.

2. **Limited Personnel:** With a small project team, responsibilities are shared across multiple roles, leading to potential communication delays when key members are unavailable or overloaded with tasks.
3. **Communication Channels:** While multiple platforms (Teams, Zoom, Messenger, Jira, OpenProject) are available, inconsistent usage and platform-switching may lead to miscommunication, missed updates, or fragmented documentation if not managed properly

Stakeholder Communication Requirements

Key stakeholders include the client, the development team, and project sponsor and adviser. Their communication preferences and needs are documented in the communications matrix and flowchart.

Roles

To support clear communication and project coordination, the following roles and responsibilities are defined for the development of the Project and Test Management System:

- **Project Sponsor / Stakeholder:** Provides overall direction and approval for the project. Reviews major deliverables and ensures the project supports the company's goals. Receives important updates on progress and outcomes.
- **Scrum Master (Project Lead):** Manages the daily activities of the team, supports smooth communication, helps solve problems, and ensures updates and concerns are shared on time. Serves as the main contact for internal coordination.
- **Product Owner:** Manages the project requirements, prioritizes features, handles the product backlog, and checks that the output meets the client's needs. Works closely with both the team and stakeholders.
- **Project Team:** Includes developers, designers, testers, and support members who complete project tasks. Each team member shares updates, reports on issues, and helps with documentation and internal communication.

Project Team Directory

Role	Name	Title	Organization/Department	Email
Project Sponsor	Prof. Bary F. Reyes	Project Sponsor	APC Faculty	baryr@apc.edu.ph
Project Manager	Hans Edwin Maybituin	Project Manager / Student	Team DHAN	hymaybituin@student.apc.edu.ph
Scrum Team (UI/UX Designer)	Hans Edwin Maybituin	Student	Team DHAN	hymaybituin@student.apc.edu.ph
Scrum Team (Developer)	Carl Agot	Student	Team DHAN	cpagot@student.apc.edu.ph
Scrum Team (QA Tester)	Noel Catabona	Student	Team DHAN	ndcatabona@student.apc.edu.ph
Scrum Master (QA Tester)	Daniel Balubar	Student	Team DHAN	

Communication Methods Technology

Team DHAN employs a range of tools to facilitate smooth communication and collaboration throughout the project. Project documents and reports are stored and managed using SharePoint, ensuring organized and secure access for all team members. GitHub is used to manage the project's codebase, track issues, and coordinate software development activities.

For task management and progress tracking, Jira and OpenProject are utilized, providing transparency and structured workflow monitoring. Official meetings and video conferences are held via Zoom, scheduled regularly to review milestones and discuss important updates with the client and the team.

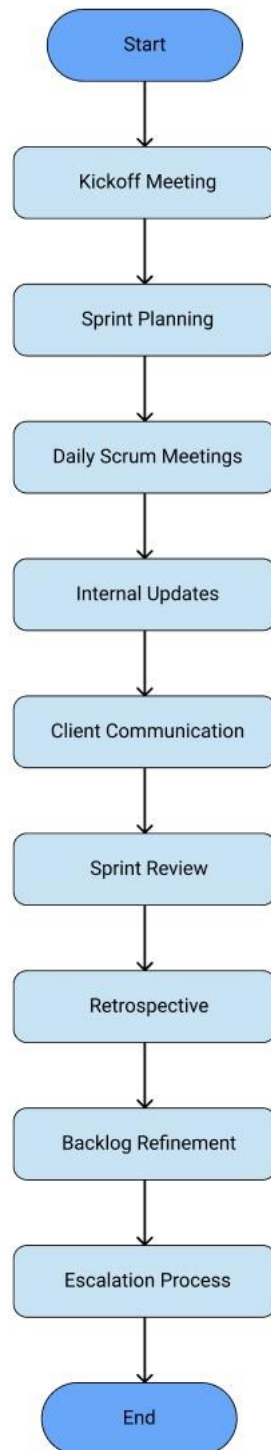
Facebook Messenger serves as the primary channel for quick, informal communication, especially for direct messaging with the client representative. Microsoft Teams is used for internal communication such as daily stand-ups and team announcements.

All participants are required to have active accounts on these platforms to maintain effective communication. The project manager keeps a record of these accounts to ensure coordination and seamless information flow.

Communication Matrix

Communication Type	Objective(s)	Medium	Frequency	Audience	Owner	Deliverable	Format
Kickoff Meeting	Introduce project scope, objectives, roles, and timeline	Microsoft Teams	Once at project start	All Stakeholders	Project Manager	Agenda, Minutes of Meeting	Soft Copy
Sprint Planning	Define sprint goals, plan tasks, and assignments	Microsoft Teams	At least twice a week	Team DHAN	Scrum Master	Agenda	MS Teams Recording Link
Sprint Review	Demonstrate completed work, gather feedback	Microsoft Teams	As needed	Team, Client	Scrum Master	Agenda	MS Teams Recording Link
Sprint Retrospective	Reflect on sprint process, identify improvements	Microsoft Teams	At least twice a week	Team DHAN	Scrum Master	Agenda	MS Teams Recording Link
Backlog Refinement	Review and update product backlog	Microsoft Teams	At least twice a week	Team DHAN	Scrum Master	Agenda	MS Teams Recording Link
Daily Scrum	Share daily progress, identify blockers	Microsoft Teams	Regularly	Team DHAN	Scrum Master	Agenda	MS Teams Recording Link

Communication Flowchart



Guidelines for Meetings

To ensure all meetings contribute effectively to project progress and stakeholder alignment, the following guidelines are established for Team DHAN:

- **Define the Purpose:** Each meeting must have a specific purpose to ensure discussions remain targeted and contribute to project goals.
- **Prepare and Share the Agenda:** A structured agenda should be prepared in advance and shared with attendees to guide the flow of conversation and prioritize topics.
- **Respect Time Commitments:** Meetings are expected to start and finish on time, acknowledging the participants' schedules and commitments.
- **Keep Records of Discussions:** Key discussion points, decisions, and task assignments are documented through meeting minutes for transparency and accountability.
- **Ensure Follow-Through:** Assigned tasks and decisions made during meetings are tracked to ensure timely execution and responsibility by the designated members.
- **Encourage Feedback:** Feedback is welcomed from participants after meetings to continually improve their structure, relevance, and effectiveness.

Communication Standards

To ensure consistent and effective communication throughout the project lifecycle, Team DHAN follows structured communication practices that support collaboration, accountability, and alignment with stakeholders.

- **Communication Protocols**

- **Meeting Frequency:** Daily check-ins via Microsoft Teams, sprint-related sessions held per cycle, and stakeholder updates scheduled as needed.
- **Preferred Platforms:** Microsoft Teams is used for internal coordination; client-facing meetings are conducted through Zoom. For informal or quick exchanges, Messenger serves as the main tool.

- **Documentation Guidelines**

- **File Formatting:** All documents follow academic formatting standards applicable to project submissions.
- **Naming System:** Files follow a consistent format such as “DHAN_ [Document Title].”
- **Storage and Versioning:** Shared folders on SharePoint are used for organized storage. Updates are tracked using date-based versioning and commit notes where applicable.

- **Reporting Practices**

1. **Content Requirements:** Regular reports include sprint outcomes, task completions, challenges encountered, and pending actions.
2. **Sharing Method:** Reports are shared through secure links in Messenger or Microsoft Teams group chats.
3. **Editing Tools:** Documents are primarily developed using Google Docs or Microsoft Word to allow real-time collaboration and accessibility.

By observing these standards, Team DIGI ensures all communication remains clear, organized, and aligned with project goals and stakeholder expectations.

Communication Escalation Process

To maintain efficient issue resolution and minimize project disruptions, Team DHAN follows a structured communication escalation process. This ensures that concerns are addressed promptly and by the appropriate level of authority.

Step 1: Identify the Issue

Team members must first recognize and document any concern that cannot be resolved within their individual or team scope—whether technical, procedural, or interpersonal.

Step 2: Notify the Scrum Master

The issue is first brought to the attention of the Scrum Master, who assesses the matter, facilitates initial troubleshooting, and determines if further escalation is necessary.

Step 3: Elevate to the Project Manager

If the concern persists or involves broader project risks (e.g., delays, resource gaps, client misunderstandings), the Scrum Master escalates the issue to the Project Manager for resolution.

Step 4: Engage the Product Owner or Stakeholder

For issues that impact project scope, timelines, or stakeholder satisfaction, the Project Manager coordinates with the Product Owner or directly consults with the client representative to align on a resolution.

Step 5: Final Escalation to Project Sponsor

In rare cases where major decisions or approvals are needed beyond the project team's authority, the issue is escalated to the Project Sponsor for final guidance or intervention.

Tracking and Documentation

Each escalation is logged through meeting notes or Jira tickets, detailing the issue, parties involved, actions taken, and the resolution. This ensures accountability and prevents recurring problems.

This escalation process promotes transparency, timely intervention, and a clear line of communication, helping Team DHAN resolve issues effectively while maintaining project momentum.

Change Management

Introduction

The Change Management Plan is essential to the successful implementation of the Project and Test Management System in ERPNext/Frappe. Since project requirements and features may change during development, this plan provides a clear process for evaluating and approving modifications. All changes will be carefully reviewed, documented, and authorized to maintain project alignment and ensure the system's successful delivery.

Change Control Board

The Change Control Board (CCB) is the primary authority responsible for overseeing all change requests related to the Team DHAN and Project and Test Management System. The CCB ensures that any proposed modifications are evaluated thoroughly to maintain alignment with the project's objectives, scope, and timeline.

Purpose of the CCB

1. **Assess Impact:** Evaluate the potential impact of proposed changes on the project's schedule, budget, and deliverables
2. **Maintain Alignment:** Ensure that all change requests support the overall project goals and stakeholder expectations
3. **Authorize Decisions:** Approve or reject change requests through a formal, transparent decision-making process

Roles and Responsibilities

Role	Description	Name
Scrum Team Member (Project Manager)	Leads the team by guiding the project, removing blockers, and making sure everyone follows the Scrum process.	Hans Edwin Y. Maybituin
Product Owner	Acts as the voice of the customer by defining project requirements, prioritizing the product backlog, and ensuring the team delivers value to the stakeholders.	Hans Edwin Y. Maybituin
Scrum Team Member (UI/UX Designer)	She is responsible for designing the user interface of the billing system and chatbot, making sure it is easy to use and visually clear.	Noel D. Catabona
Scrum Team Member (Developer)	He is responsible for developing the system features, writing the code, and ensuring all parts of the billing system and chatbot function properly.	Carl P. Agot
Scrum Team Master (Quality Assurance)	He is responsible for testing the system, finding bugs, and helping to make sure the KPI's and real time data work correctly and reliably.	Noel D. Catabona
Scrum Team Member (Quality Assurance)	He is responsible for checking and reviewing the system, confirming that any issues are fixed and that the product meets the expected quality.	Daniel Louis Balubar
Stakeholder	Provides key input on project scope, requirements, and priorities. Approves the system's readiness for deployment and supports the transition from development to implementation.	Prof. Bary Reyes / Asia Pacific College - Students
Class Adviser	Offers guidance and support to the Project Manager and team, ensuring the project aligns with academic standards and requirements.	Prof. Jose Eugenio Quesada
Project Adviser	Provides technical and strategic advice to the team, ensuring that the project delivers value and meets the objectives agreed upon by the Product Owner and Scrum Team.	Prof. Felino Calderon

Change Control Process

The following standardized process will be used for managing all change requests in the Subscriber Billing System with an AI-Powered Chatbot project:

Change Request Submission

- Any team member or stakeholder can submit a Change Request Form through platforms such as Jira or email.
- The form must include a description of the change, the reason for the request, its urgency, and the expected impact on the project.

Initial Review

- The Project Manager reviews the submitted request for completeness and relevance.
- Valid requests are logged and forwarded to the Change Control Board (CCB) for evaluation.

Impact Analysis

1. The Scrum Team, Product Owner, and any relevant technical specialists (e.g., for UI/UX or AI features) assess the change.
2. The team evaluates the impact on cost, timeline, technical requirements, and project goals.
3. The results of the analysis are shared with the CCB to support decision-making.

CCB Decision

- The CCB reviews all findings and votes to approve, reject, or defer the change.
- Decisions and their justifications are documented and communicated to all relevant parties.

Implementation (if approved)

- Once approved, the Project Manager updates the backlog, project plan, and documentation.
- The change is assigned to the appropriate sprint or development cycle.

System Testing

- The QA team or designated testers verify that the implemented change functions correctly within the integrated system.
- This includes validating both the new feature and fix and ensuring existing functionality is not negatively impacted.
- Any defects found are logged, addressed, and retested before moving forward.

Monitoring

- The Scrum Master ensures that the implemented change is functioning as intended.
- The change is monitored to prevent regression and ensure alignment with project objectives.

Documentation

- All relevant technical documents, user guides, and change logs are updated to reflect the newly implemented change.
- This ensures clarity for end-users, support teams, and future developers, contributing to long-term maintainability and knowledge transfer.

Communication

- All stakeholders are notified of the decision and any updates to the project plan.

- Supporting tools and repositories (e.g., Jira, OpenProject, SharePoint) are updated accordingly.

Implementation/Transition

Executive Summary

This transition-out plan defines the strategy and actions required to smoothly disengage from the development and implementation phase of the project and test management system. The primary objective is to ensure business continuity, data security, and maintainable system performance.

Transition Team Organization

Name	Role
Hans Edwin Y. Maybituin	Project Manager
Carl P. Agot	Developer
Noel D. Catabona	QA Tester
Daniel Louis Balubar	QA Tester

Workforce Transition

Team DHAN will remain in contact with the client. They are available for extended consultation or future feature requests as needed through a freelance or contract agreement.

Work Executing During Transition

Work during the transition includes:

- Access validation
- Bug fixing and post-deployment patches
- Performance monitoring

No major development tasks will be ongoing but monitoring and optimization tasks will be executed.

Property Transition

Intellectual Property

All developed source code, design documents, and training materials will be transferred to the client (Bary F. Reyes).

User Accounts and Passwords

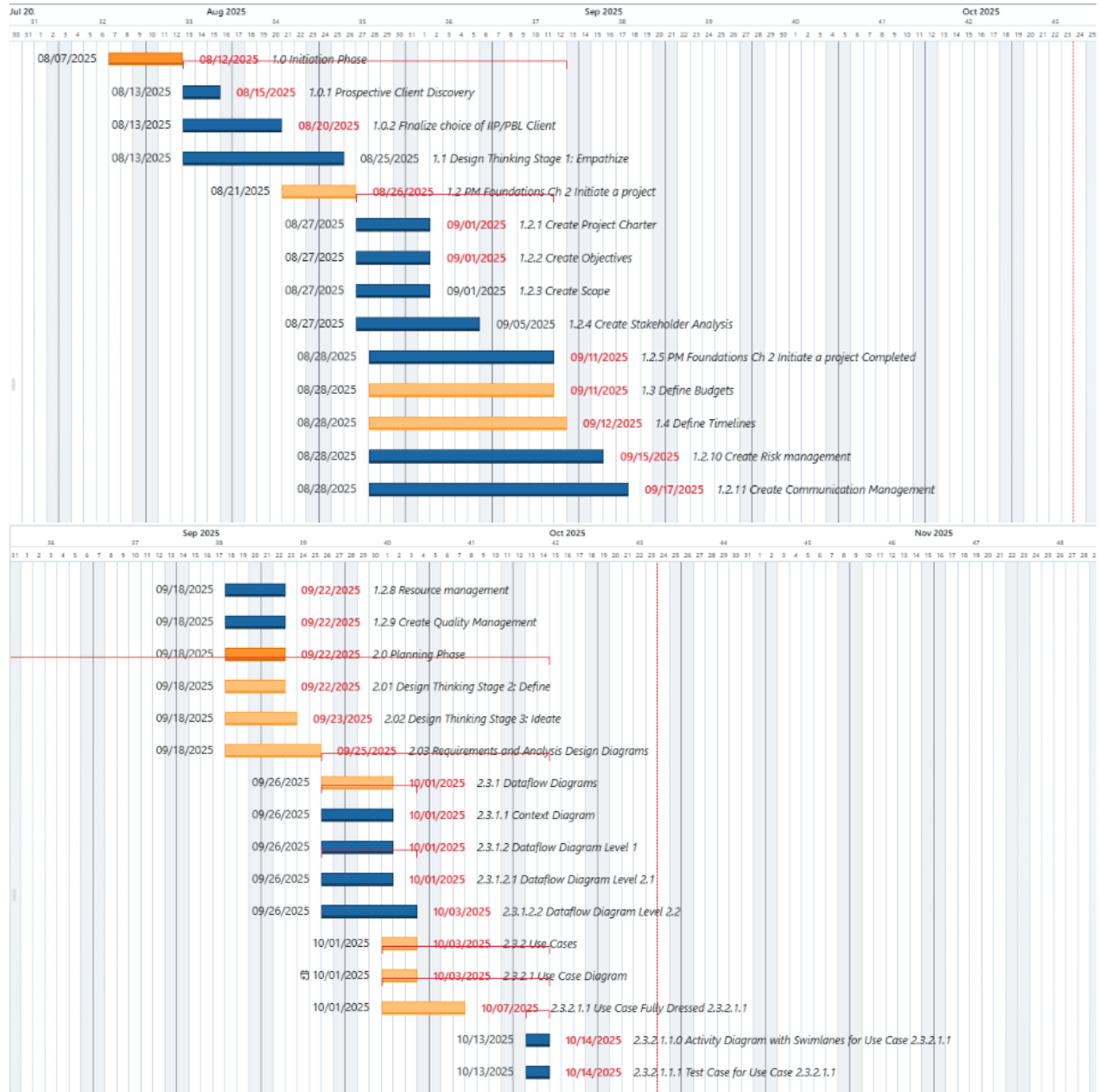
Account Type	Roles	Description
Test_admin	Project Manager	Has access to all features and all capabilities of the system
Test-dev	Developer	Can report an issue, select a project, and update tasks.
Test- tester	Tester	Report an issue, create and update a test case and test suite.

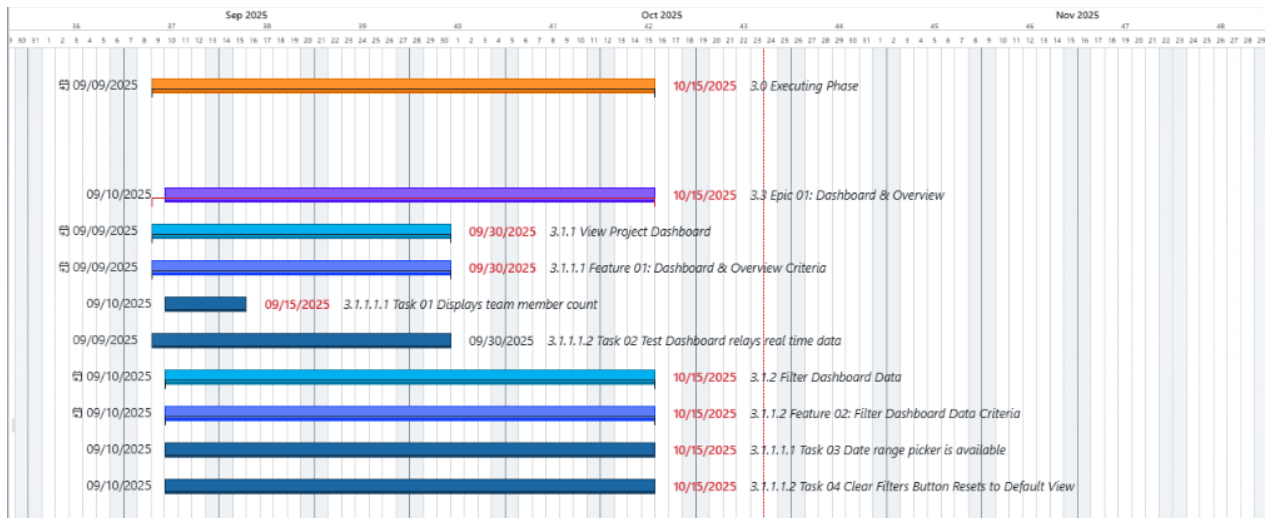
Knowledge Transfer

Knowledge will be transferred through:

- Transfer of manuals (admin, end-user, and system manual)
 - Access to a recorded training video and system demo
- Email/Slack channel maintained for 30 days post-handover

Schedule





This Gantt chart presents the DHAN Project and Test Management system's structured timeline. It spans from Aug. 2025 to late 2025, detailing phases such as Initiation, Stakeholder Engagement, Project Planning, and Execution (including Design, Development, and Implementation). Tasks are visually organized with color-coded bars—green for task durations and orange for key deadlines—along with arrows indicating task dependencies. Major milestones

include the Business Case, Charter Development, and System Testing, ensuring a clear path from planning to deployment and quality assurance.

Handover Acceptance

The project team will facilitate an acceptance review meeting with:

- Project Adviser
- Project Client

A formal checklist-based sign-off will be executed, including:

- Confirmation of system stability
- Verification of access and credentials
- Acceptance of knowledge transfer materials